



CORRECTIN: LANGUAGE EXCHANGE APP

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LIST OF ABBREVIATIONS

i.e.	Id est (Latin: this means)
e.g.	Exempli gratia (Latin: for example)
NSPE	National Society of Professional Engineers
IEEE	Institute of Electrical and Electronics Engineers
CPU	Central Processing Unit

1 Realistic Constraints

1.1 Social, Environmental and Economic Impact

For a project to be considered a sustainable development, it should have positive social, economical and environmental impacts.

It is a cost-effective language learning application for users who will use the system socially and economically, and it offers the opportunity to meet and interact with new people from all over the world socially. Along with the new features to be added to the application, it is a source of income for language instructors who will start using the application, and at the same time, it constitutes a possible income source model for companies that want to sponsor and invest in the application.

Being able to create social benefits for the public, creating demand economically, creating resources to finance itself, and not being environmentally harmful are important key points. The application, which was developed by taking these key points into account, aims to provide added value to the state with income tax payments when the application reaches a position that appeals to large masses. It is expected to create an interactive social effect that people can interact with as a new social media platform when it is achieved to use a large mass in the social sense.

1.2 Cost Analysis

In this section, we will talk about the cost spent to carry out this study and the costs that may arise in possible cases.

Web hosting services are needed for the application to reach the end user. In line with the approach recommended in the industry, it is a safe and recommended solution to have the frontend, backend and database on separate servers. In line with this scope, hosting and domain name are required to deploy the front of the project, and backend server and database server are needed to deploy the backend application. Since it is aimed to keep the current needs at a minimum level, we determined the server features at the lowest level and made a price analysis. In the future, if our application receives heavy traffic, it may be necessary to rent extra servers and load balancing so that the system can continue to work stably.

When the minimum expenses were analyzed at the first stage, the prices of the turkey-based natro company [1] as a cloud service provider were examined and the 1-year expenses are as follows:

- Domain name (www.correctin.com.tr) : 33,22 TL (year)
- Hosting (1 Web Page, 10GB Disk, 500GB Traffic : 158,28 TL (year)
- Backend Server (1CPU, 1GB Memory, 20GB SSD) : 779,18 TL (year)
- Database Server (1CPU, 1GB Memory, 20GB SSD) : 779,18 TL (year)
- Total : 1749,86 TL

If we take as a reference that a newly graduated computer engineer in Turkey works for minimum wage and taking into account that we developed the project as 2 people and the project was completed after 3 months of work, total employee cost :

- $4250 * 3 \text{ (month)} * 2 \text{ (developer)} : 25,550 \text{ TL}$

If we consider the processes of continuing the development and maintenance of the project until the end of the year, an additional effort of about 5 months is required:

- $4250 * 5 \text{ (month)} * 2 \text{ (developer)} : 42,500 \text{ TL}$
- Total 1 year expense: 69.749,86 TL

1.3 Standards

There are some standards in that engineers should follow or meet its requirements to perform a trustworthy and ethical project. In this section two standards NSPE Code of Ethics and IEEE Standard will be investigated.

1.3.1 NSPE Code of Ethics

Engineering is a vital and complex profession. As members of the profession, engineers are expected to uphold the highest standards of honesty and integrity. Engineering has a direct and profound influence on every person's quality of life. Therefore, engineers must be dedicated to upholding the general public's health, safety, and welfare and must perform their services with honesty, objectivity, justice, and equity. A professional code of conduct that demands adherence to the highest moral standards is applied to engineers. [6]

1. Fundamental Canons

Engineers, in the fulfillment of their professional duties, shall:

- Hold paramount the safety, health, and welfare of the public.

2. Rules of Practice

- Engineers shall hold paramount the safety, health, and welfare of the public.
 - Engineers shall approve only those engineering documents that are in conformity with applicable standards.
- Engineers shall issue public statements only in an objective and truthful manner.
 - Engineers may express publicly technical opinions that are founded upon knowledge of the facts and competence in the subject matter.

3. Professional Obligations

- Engineers shall be guided in all their relations by the highest standards of honesty and integrity.
 - Engineers shall advise their clients or employers when they believe a project will not be successful.

1.3.2 IEEE Standards

- ISO/IEC/IEEE Draft International Standard - Systems and software engineering—Systems and software assurance—Part 2: Assurance case
 - This document specifies minimum requirements for the structure and its meaning of assurance cases. It does not place requirements on the quality of the contents but describes the structure and its meaning of assurance cases with the necessary level of precision and detail so as to avoid inconsistent and subjective use of the terms. While several notations and slightly varying terminologies are currently used in practice, this document does not require the use of any particular concrete representation including graphical representation. Likewise, it places no requirements on the means of physical implementation of the data; in particular, it includes no requirements for redundancy or co-location.

2 Risk Analysis and Precautions

In today worlds, with the rapid progress of technology, possible new risks are increasing day by day. In this section, we discussed the possible risks for our project.

- Risk 1 [2] : In our application, we made the necessary validations in the possible create and update processes, both on the front and on the back side, and tested whether we got the correct results, but we doubt the controls that are likely to be overlooked in this application created with manpower.
- Risk 2 [3]: We have ensured the security of the API by taking security measures against possible attacks against our API, but since we did not have a cybersecurity expert perform penetration tests on our API, we are suspicious of possible penetration attempts.
- Risk 3 [4] : We kept critical user information such as passwords in our database in an encrypted and undecipherable manner by asking the user to enter a secure password, and we did not transfer this information over the network when returning user information to the client side, but we suspect a possible data leak since we did not have it tested by cyber security experts.
- Risk 4 [5] : We did not perform load and stress tests on our API, so we cannot predict the maximum amount of user traffic it can handle.

3 Contributions

Students in the project are Berk Özpinar and Muhammed Sedef.

Berk Özpinar (FrontEnd and Mobile Development) :

- Creating the frontend of the web application using Angular (Type-script), HTML and CSS.
- Creating the cross platform mobile application using Flutter (Dart).
- Designing the both web and mobile application.
- Working with project teammate Muhammed Sedef to create the project report and presentation.
- Implementing and testing API connections.

Muhammed Sedef (BackEnd and Database Development) :

- Designing the database architecture required for use in the application
- Creating appropriate relational database model and tables according to the designed database architecture
- Analyzing the services needed by the client side in the application and developing RESTFUL API using the appropriate layered architecture
- Backend coding with java spring boot framework in a simple, understandable, solid principles and open to development and closed to change
- Working with his teammate Berk Özpinar to create a project report and presentation
- Execution of the deployment processes of the backend and database of the project

4 Conclusion

A cost effective language learning application for language learners and teachers has been successfully implemented, the application offers opportunity to meet and interact with the new people from all over the world socially. Also the teachers in the future can possibly grant incomes from the application and at the same time it forms a possible income source model for companies that want to sponsor and invest in the application.

The application's frontend, backend and database are deployed onto separate servers to follow industry standards. Considering this a yearly cost to have the application running was estimated to be around 1749.86 TL, excluding the costs for wages of the developers. If developer wages to be calculated from minimum wages, it would sum up to 25.550 TL for 2 developers over 3 months period. In total, the application has been developed and deployed for 27.299 TL.

While developing, NSPE Code of Ethics and IEEE standards have been followed for design and deployment of the application. Increasing the welfare of the public is one of the main goals of the application, which also corresponds with the fundamental canons of the NSPE Code of Ethics. Furthermore, in order to provide a security system with industry standards ISO/IEC 20734-1:2011 ¹ was also considered.

Overall, a socially and economically positive, cost effective application had been developed, the application also meets the requirements of various software development standarts.

¹ISO/IEC 27034-1:2011 Information technology, security techniques, application security

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